GeoClaw-Claude

User Guide v3.2.0

Natural Language-Driven Intelligent GIS Platform

UrbanComp Lab | China University of Geosciences (Wuhan) | urbancomp.net

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1. Installation & Setup

1.1 Prerequisites

- Python 3.9 or later
- Git
- At least one LLM provider: Anthropic Claude / Google Gemini / OpenAI GPT / Qwen API key, or a local Ollama installation

1.2 Installation

```
git clone https://github.com/whuyao/GeoClaw_Claude.git
cd GeoClaw_Claude && bash install.sh
```

For development installation with editable mode:

```
bash install.sh --dev
```

1.3 Initial Configuration (onboard)

After installation, run the two-step onboard wizard:

```
geoclaw-claude onboard
```

The wizard will ask you to: (1) select your LLM provider and enter API key, (2) set the output directory for generated maps and analysis results.

1.4 Advanced Configuration

```
geoclaw-claude config
```

This interactive menu allows you to modify proxy settings, DPI, log level, cache location, and other advanced parameters.

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2. Basic Usage

2.1 Single Query Mode

Run a one-shot analysis query:

```
geoclaw-claude ask "下载武汉市医院数据，做 1 公里缓冲区，用交互地图显示"
geoclaw-claude ask "Load hospital data for Wuhan, create 1km buffers, show interactive map"
```

2.2 Multi-Turn Chat Mode

Start an interactive session with conversation history:

```
geoclaw-claude chat
```

In chat mode, you can use pronoun-like references such as 'now clip it to the district boundary' and the system will resolve them from session context.

2.3 Common Workflows

Data loading and basic analysis:

```
ask "加载 hospitals.geojson 文件"
ask "对医院做 500 米缓冲区"
ask "把缓冲区裁剪到武汉市范围内"
ask "用静态地图显示结果"
```

Network analysis:

```
ask "计算从汉口火车站出发 15 分钟步行等时圈"
ask "分析从 A 点到 B 点的最短路径"
```

Raster analysis:

```
ask "分析 dem.tif 的坡度和坡向"
ask "对 landuse.tif 进行重分类：1-3 类归为低密度，4-6 类归为高密度"
```

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3. Skill System

3.1 Running Built-in Skills

Use the skill run command to execute any of the 15 built-in Skills:

```
geoclaw-claude skill list                              # Show all available Skills
geoclaw-claude skill run vec_buffer --distance=500    # Buffer analysis
geoclaw-claude skill run retail_site_algo --data candidates.geojson --top_n=5
```

```
geoclaw-claude skill run hospital_coverage --hospitals hospitals.geojson --city
wuhan
```

3.2 Installing Custom Skills

```
geoclaw-claude skill install ./my_analysis.py
```

The Skill file must define SKILL_META and a run(ctx) function. See the Skill Writing Guide for the complete specification.

3.3 Skill Security Audit

Before running any third-party Skill, audit it with:

```
geoclaw-claude skill audit my_analysis
```

The auditor reports risk levels from INFO through CRITICAL. Always review CRITICAL and HIGH findings before running.

3.4 Exporting Skills to OpenClaw (v3.2.0)

Export any GeoClaw Skill as an OpenClaw / AgentSkills-compatible SKILL.md:

```
geoclaw-claude skill export vec_buffer                # Export single Skill
geoclaw-claude skill export --all                     # Export all built-in
Skills
geoclaw-claude skill export --all --output ./export/  # Specify output
directory
```

The exported SKILL.md can be installed directly into OpenClaw. This allows OpenClaw users to leverage GeoClaw's GIS analysis capabilities.

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4. Memory System

4.1 Storing Knowledge

```
geoclaw-claude memory add "武汉市人口重心在汉口片区" --tag 武汉 --importance 0.8
```

4.2 Searching Memory

```
geoclaw-claude memory vsearch "城市人口分布"   # Vector semantic search
geoclaw-claude memory search "武汉"           # Keyword search
geoclaw-claude memory recent --limit 5        # Most recent entries
```

4.3 Memory Auto-Debrief

At the end of every chat session, the system automatically generates a debrief entry recording: complete operation sequence, execution statistics, intermediate results flagged for retention, and session insights inferred from conversation.

4.4 Enabling Neural Vector Search

For higher recall on semantic queries, install the optional sentence-transformers package:

```
pip install sentence-transformers
```

GeoClaw will automatically detect and use it for the vsearch command.

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5. Personalization (soul.md & user.md)

5.1 Files Location

Both files are stored in ~/.geoclaw_claude/:

- soul.md — System identity, analysis principles, safety boundaries (partially locked)
- user.md — User preferences, inferred research domain, session insights (auto-updated)

5.2 Viewing and Editing

```
geoclaw-claude profile view soul     # View soul.md
geoclaw-claude profile view user     # View user.md
geoclaw-claude profile edit user     # Edit user.md manually
```

5.3 Automatic Updates

user.md is automatically updated at the end of each chat session. The system infers:

- City preferences — cities mentioned 2+ times in session
- Research domain — keyword-based scoring across 8 domain categories
- Session insights — timestamped summary appended each session
- Language and style preferences — inferred from message character ratios and length

5.4 Safety Locks

The following soul.md sections cannot be modified through conversation:

Locked Section	Protection Reason
Safety Boundaries	Prevents disabling file access restrictions and data exfiltration guards
Execution Hierarchy	Maintains tool priority order to prevent misuse
Core Principles	Preserves correctness / reproducibility / transparency commitments
Data Handling Rules	Enforces read-only data access and output path restrictions

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6. Multi-LLM Provider

6.1 Cloud Providers

Provider	Setup Command	Notes
Anthropic Claude	geoclaw-claude onboard -> select anthropic	Best NL understanding, recommended for complex GIS queries
Google Gemini	geoclaw-claude onboard -> select gemini	Fast, good Chinese support, free tier available
OpenAI GPT	geoclaw-claude onboard -> select openai	Broad compatibility
Qwen (Alibaba)	geoclaw-claude onboard -> select qwen	Chinese-optimized, OpenAI-compatible API

6.2 Ollama (Local / Offline)

For completely offline use without an API key:

```
# Install Ollama from https://ollama.com/download
ollama serve
ollama pull qwen3:8b     # Recommended default (6 GB VRAM)
geoclaw-claude onboard   # Select ollama, enter model name
```

For team / LAN deployment:

```
# Edit ~/.geoclaw_claude/config.json
{ "llm_provider": "ollama", "ollama_base_url": "http://192.168.1.100:11434/v1",
"ollama_model": "qwen3:14b" }
```

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7. Spatial Reasoning Engine (SRE)

SRE automatically reasons about your analysis query before execution. You can inspect the reasoning output with:

```

from geoclaw_claude.reasoning import reason
result = reason("分析武汉地铁站周边商业活跃度",
                datasets=[{"id": "metro", "type": "vector", "crs":
"EPSG:4326"}])
print(result.summary_text(lang='zh'))
```

Key fields in the reasoning result:

- uncertainty_score (0-1) — Overall analytical uncertainty with 5-dimension breakdown
- analysis_mode — exploratory / confirmatory / causal / descriptive
- parameter_sensitivity — Sensitivity and suggested ranges for key parameters
- maup_risk — Risk of Modifiable Areal Unit Problem

- workflow_steps — Ordered recommended analysis steps

Note: When analysis_mode is 'causal', SRE automatically generates a warning that GIS analysis alone cannot establish causality. Consider supplementing with GWR, DID, or other methods.

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8. CLI Reference

Command	Description
geoclaw-claude ask <query>	Single natural language query
geoclaw-claude chat	Start multi-turn interactive session
geoclaw-claude onboard	Initial configuration wizard
geoclaw-claude config	Modify configuration interactively
geoclaw-claude skill list	List all registered Skills
geoclaw-claude skill run <name> [params]	Execute a Skill
geoclaw-claude skill install <file>	Install a Skill from Python file
geoclaw-claude skill audit <name>	Security audit a Skill
geoclaw-claude skill export <name>	Export Skill to OpenClaw format (v3.2.0)
geoclaw-claude skill export --all	Batch export all Skills (v3.2.0)
geoclaw-claude memory add <text>	Add entry to long-term memory
geoclaw-claude memory search <query>	Keyword search memory
geoclaw-claude memory vsearch <query>	Vector semantic search memory
geoclaw-claude memory recent	Show recent memory entries
geoclaw-claude profile view <soul|user>	View personalization file
geoclaw-claude profile edit <user>	Edit user.md
geoclaw-claude check	Check for newer version
geoclaw-claude update	Pull latest and reinstall
geoclaw-claude self-check	Full health check report

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Web 对话界面

GeoClaw v3.2.0 新增了浏览器对话界面，无需终端即可与框架交互。界面采用深色地图科技感主题，坐标格背景，水绿色强调色，配以 Noto Serif SC + JetBrains Mono 字体。

启动方式

```
pip install flask flask-cors
python web/server.py                    # 默认 http://localhost:7860
python web/server.py --port 8080        # 自定义端口
python web/server.py --rule             # 离线规则模式，无需 API Key
```

启动后打开浏览器访问提示的地址即可使用，用户数据（配置、记忆）完全不受影响。

界面功能说明

侧边栏：实时显示 AI 引擎、Provider、会话 ID、图层数量；7 个示例指令一键填充；底部版权信息（UrbanComp Lab · 中国地质大学（武汉）· GitHub · MIT License）。

对话区：多轮对话，动态气泡动效，打字指示符，代码块高亮，每条回复显示 Provider 与耗时。

顶栏：会话 ID 徽章，浮动图层面板（实时列出已加载 GeoLayer），新建会话按钮。

开发：UrbanComp Lab · 中国地质大学（武汉）· github.com/whuyao/GeoClaw_Claude · © 2025 MIT License